

OBIHIRO, Japan

WMO: 474170

Lat: 42.9222N

Lon: 143.2119E

Elev: 144

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	3.3	-6.2	4.0	1.8	-3.0	4.7	6.1	20.5	28.4	18.6	28.3	3.2	210	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	86.9	71.8	83.3	70.2	79.7	68.3	74.6	83.3	72.7	79.8	70.8	76.8	7.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	118.7	78.5	70.3	112.2	76.6	68.7	105.9	74.5	38.3	83.7	36.5	79.3	34.9	77.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.0	13.7	-6.2	92.9	3.8	3.1	-9.0	95.2	-11.2	97.0	-13.3	98.8	-16.0	101.0		
			-6.8	77.7	3.7	2.1	-9.4	79.3	-11.6	80.5	-13.7	81.7	-16.4	83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	18.8	21.3	31.3	43.5	53.7	60.6	67.4	69.7	63.5	51.0	38.5	24.8
	DBStd	18.61	6.41	6.74	6.42	6.37	6.56	5.99	5.94	5.85	5.80	5.74	6.50	6.34
	HDD50	3790	967	805	581	214	36	2	0	0	0	57	348	780
	HDD65	7494	1432	1225	1046	645	354	156	44	21	97	433	796	1245
	CDD50	2136	0	0	0	19	151	321	539	611	404	89	2	0
	CDD65	366	0	0	0	0	4	25	118	167	52	0	0	0
	CDH74	2431	0	0	0	9	120	270	787	992	249	4	0	0
	CDH80	723	0	0	0	1	30	64	251	332	45	0	0	0
Wind	WSAvg	4.8	5.2	5.5	5.8	6.0	5.0	4.0	3.5	3.4	3.7	4.6	5.4	5.4
Precipitation	PrecAvg	39.3	1.7	1.3	1.8	2.6	3.7	3.3	4.5	5.9	6.0	3.6	2.6	2.4
	PrecMax	53.2	4.2	3.3	5.0	10.4	8.0	9.6	10.4	15.5	13.9	7.0	6.7	5.9
	PrecMin	22.1	0.3	0.2	0.3	0.4	1.2	1.3	1.2	2.5	2.5	0.7	0.3	0.6
	PrecStd	6.7	1.2	0.9	1.2	2.0	1.9	2.1	2.3	2.8	2.7	1.9	1.7	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	42.2	55.4	74.0	83.6	86.4	91.0	92.7	85.0	71.9	61.2	45.6
		MCWB	31.9	35.1	43.4	52.9	60.4	67.0	73.7	76.2	69.7	58.7	50.7	39.7
	2%	DB	34.9	38.4	49.2	66.1	76.8	81.0	86.2	87.6	80.1	67.5	56.4	40.5
		MCWB	30.2	32.3	39.6	50.8	57.7	65.0	72.2	73.7	68.4	56.4	48.0	35.1
	5%	DB	32.6	35.3	44.7	60.7	71.7	76.6	82.5	83.6	76.6	64.0	52.8	37.5
		MCWB	28.2	30.0	36.8	47.4	56.0	64.0	70.8	72.0	65.9	55.0	45.8	32.7
	10%	DB	30.4	32.8	41.3	55.8	66.7	72.1	78.1	79.6	73.1	61.1	49.1	34.9
		MCWB	26.4	28.3	34.5	45.2	54.0	61.9	69.3	70.3	64.8	53.4	43.4	30.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	35.8	44.7	55.5	63.5	69.6	76.1	77.9	74.1	62.4	53.2	40.6
		MCDB	35.9	41.1	53.7	67.7	79.6	80.8	85.6	88.2	79.8	66.5	57.1	44.2
	2%	WB	31.0	32.9	40.5	51.9	60.1	67.1	74.0	75.6	71.3	59.2	49.6	35.9
		MCDB	34.0	37.3	47.5	64.0	71.9	78.0	83.0	85.3	76.2	64.4	54.3	39.5
	5%	WB	28.7	30.7	37.7	48.9	57.7	64.7	72.0	73.5	68.9	57.1	46.7	33.2
		MCDB	32.1	34.6	43.7	58.6	68.3	74.3	80.4	80.6	73.6	61.4	51.4	36.6
	10%	WB	26.6	28.7	35.1	46.2	55.6	62.6	70.0	71.6	66.5	55.0	43.9	31.1
		MCDB	30.2	32.6	40.3	54.2	64.8	70.2	76.6	77.1	71.0	59.3	48.5	34.4
Mean Daily Temperature Range	5% DB	MDBR	19.5	19.6	16.8	19.1	19.7	17.7	15.4	14.4	15.6	17.7	15.7	16.7
		MCDBR	18.1	19.7	21.0	28.6	30.0	26.5	23.5	21.1	20.2	21.1	21.5	16.8
		MCWBR	14.5	15.2	13.5	15.3	14.5	13.5	11.7	10.1	10.2	12.4	14.3	12.7
	5% WB	MCDBR	16.8	18.4	20.2	25.7	25.1	23.3	20.8	18.4	16.4	17.2	19.6	15.9
MCWBR		14.1	14.6	13.8	14.9	13.7	12.8	11.3	9.7	9.6	11.3	14.1	12.6	
Clear-Sky Solar Irradiance	taub		0.318	0.370	0.450	0.508	0.526	0.505	0.489	0.469	0.411	0.394	0.367	0.329
	taud		2.083	1.917	1.804	1.814	1.868	2.008	2.122	2.194	2.313	2.230	2.207	2.136
	Ebn at Noon		248	251	245	242	242	248	251	252	259	245	230	232
	Edh at Noon		33	47	60	64	62	55	48	43	36	34	30	29
All-Sky Solar Radiation	RadAvg		598	873	1222	1476	1569	1555	1418	1300	1131	892	593	486
	RadStd		38	61	87	116	134	136	142	98	96	53	42	41

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.56	N/A	N/A	+1.49	N/A	N/A	N/A	-171	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page